Neel Kochhar

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EDUCATION

University of Calgary — Bachelor of Science in Computer Science September 2026 – Present

Academic Focus: Computer Science with a strong interest in machine learning and mathematics.

TECHNICAL SKILLS

Languages & Tools: Python, JavaScript, C++, SQL, HTML, Docker, Git, GitHub, Unix Shell, pytest, Raylib
Backend & LLM Engineering: FastAPI, Pydantic, SQLite, REST APIs, Anthropic API (tool use), structured LLM output (Instructor), prompt engineering, LLM evaluation
Deep Learning & AI: PyTorch, NumPy, Transformers, Self-Attention, CNNs, RNNs, Autoencoders, Backpropagation
Mathematics: Calculus, Linear Algebra

MACHINE LEARNING AND SOFTWARE PROJECTS

Cadence — Natural-Language Calendar Agent | Python, FastAPI, Anthropic API, SQLite, JavaScript, Docker Live Demo • GitHub

- Built and deployed (Docker, Fly.io) a calendar app where plain-English requests become events. Quick mode turns one sentence into a validated add/update/remove (Pydantic + Instructor, swappable Claude/Ollama provider); agent mode handles multi-step requests like “cancel everything on Saturday and add a hike at 9am”.
- Wrote a tool-calling agent loop on the Anthropic SDK (search/add/update/remove tools, 8-step cap) that feeds validation errors back to the model so it self-corrects; the session ID is never in a tool schema, so the model cannot reach another visitor’s data.
- Added conflict detection that holds overlapping events for user confirmation (the agent has no confirm tool, so it cannot approve its own conflicts), plus per-visitor sessions and IP rate limiting. Fixed LLM date errors by giving the model a 7-day lookup table instead of asking it to do date math.
- Built state-based evals that run the real model and check the final calendar: 59 hand-written cases, 100% pass on Claude Haiku (50 quick-mode cases at ~1.3–1.8 s avg latency; 9 agent cases at 3.45 s avg, ~2.1 tool steps). Plus 55 pytest tests. Wrote the vanilla JavaScript/HTML frontend (FullCalendar).

Deep Learning Architectures from First Principles | PyTorch, Python GitHub

- Implemented deep learning models from scratch in PyTorch, including Transformers with multi-head self-attention and Autoencoders.
- Validated model performance and convergence on benchmark datasets, including CIFAR-10 for image classification and SST-2 for sentiment analysis.

Mathematical Foundations of Machine Learning | Python, NumPy GitHub

- Built a multi-layer neural network using strictly NumPy (no autograd) to solve non-linearly separable classification (XOR); derived and coded forward propagation, cross-entropy loss, and manual backpropagation.
- Authored a research paper on gradient descent analyzing loss surface topology and convergence behavior.

Interactive Pendulum Physics Engine & Simulation | C++, Raylib GitHub

- Built a real-time 2D pendulum simulation in C++ modelling gravity and damping, using delta-time frame scaling for angular acceleration and velocity, polar-to-Cartesian mapping, and an object-oriented design with interactive mouse-drag.

LEADERSHIP & ATHLETIC ACHIEVEMENT

National-Level Karate Athlete (U21 –60 kg) — WKF Kumite 2023 – Present

- Compete in high-performance WKF kumite at national-level tournaments while maintaining a daily conditioning and training schedule alongside full-time studies.

Assistant Karate Instructor & Dojo Volunteer 2023 – 2026

- Completed 600+ volunteer hours instructing junior athletes in technique, conditioning, and kumite tactics; assisted head instructors with dojo operations and event organization.